Topic 1: Generalized Anxiety Disorder (GAD) & Acute Agitation

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Dysregulation in amygdala and frontolimbic circuits involving serotonin, norepinephrine, and GABA neurotransmission. GAD defined as excessive, uncontrollable anxiety and worry occurring more days than not for at least 6 months about a number of events or activities.

CLINICAL PRESENTATION & PHYSICAL EXAM

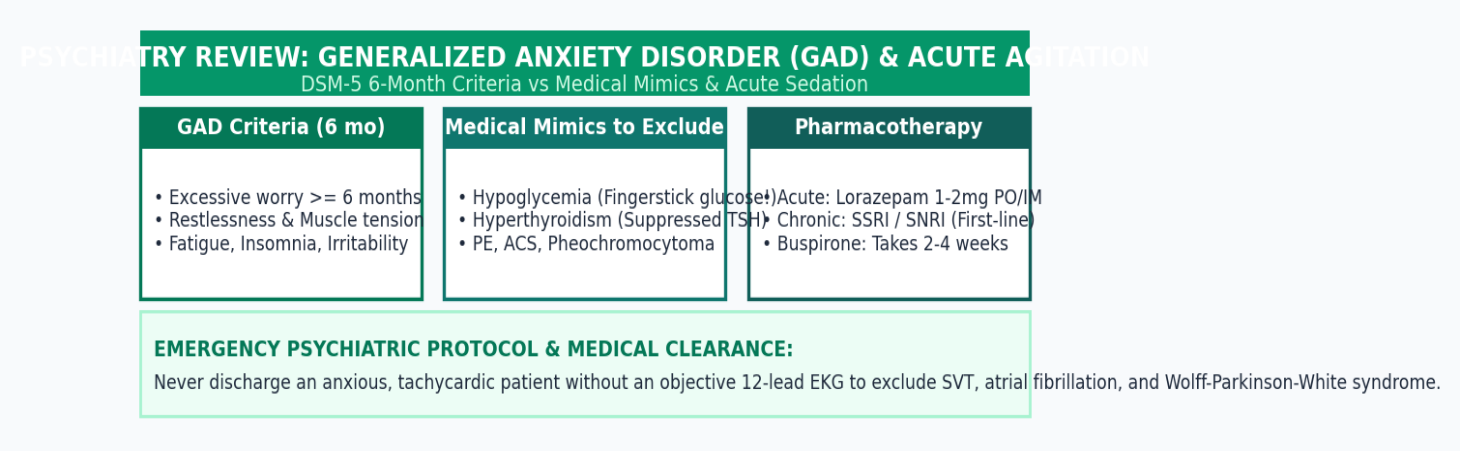
• **Presentation:** Excessive worry associated with at least 3 of 6 symptoms: restlessness/feeling keyed up, easy fatigability, difficulty concentrating, irritability, muscle tension, and sleep disturbance.
• **Physical Exam:** Tremulousness, diaphoresis, tachycardia, tachypnea, motor agitation. Complete neurological exam normal.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Diagnosis of exclusion in the emergency department. Rule out medical mimics: Hypoglycemia, Hyperthyroidism, Pheochromocytoma, Pulmonary Embolism, Acute Arrhythmias, and Sympathomimetic toxicity (Cocaine, Methamphetamine). EKG, Fingerstick glucose, TSH, urine tox screen.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Acute Severe Agitation: Low-dose short-acting oral Benzodiazepine (Lorazepam 1-2 mg PO/IM). Long-term First-Line: SSRIs (Escitalopram, Sertraline) or SNRIs (Venlafaxine) + Cognitive Behavioral Therapy (CBT). Buspirone (5-HT1A partial agonist) for chronic GAD (no sedation/abuse potential, but takes 2-4 weeks to take effect).



HIGH-YIELD BOARD PEARL: Medical mimics must always be excluded before diagnosing primary anxiety: check fingerstick glucose, 12-lead EKG, and thyroid function. Buspirone is non-sedating and lacks addiction potential, but is ineffective for acute anxiety attacks due to a delayed onset of 2-4 weeks.

Topic 2: Bipolar Disorder & Acute Mania

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Chronic mood disorder. Bipolar I: at least ONE lifetime MANIC episode (lasting >= 7 days or requiring hospitalization; severe functional impairment). Bipolar II: at least one HYPOMANIC episode (>= 4 days, NO marked functional impairment, NO psychosis) AND at least one Major Depressive Episode.

CLINICAL PRESENTATION & PHYSICAL EXAM

• **Presentation:** Manic episode mnemonic "DIG FAST": Distractibility, Impulsivity/Indiscretion (spending, hypersexuality), Grandiosity (inflated self-esteem), Flight of ideas/racing thoughts, Activity increased (goal-directed), Sleep need decreased (sleeps 2h, feels fully rested!), Talkativeness (pressured speech).
• **Physical Exam:** Pressured rapid speech, psychomotor agitation, grandiose delusions, auditory hallucinations, poor insight, hyperactive motor behavior.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Clinical psychiatric evaluation. Comprehensive medical workup: Toxicology screen, Head CT (first-onset mania in elderly), TSH, CMP. Rule out steroid-induced mania and stimulant toxicity.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Acute Manic Agitation in ED: Second-generation Antipsychotic (Haloperidol 5 mg IM OR Olanzapine 10 mg IM OR Ziprasidone 20 mg IM) PLUS Benzodiazepine (Lorazepam 2 mg IM) for rapid tranquilization. Mood Stabilizers for maintenance: Lithium (narrow therapeutic window 0.8-1.2 mEq/L; nephrotoxic, check level Q3-6 mo), Valproic Acid / Divalproex, or Atypical Antipsychotics (Quetiapine, Aripiprazole). Antidepressant monotherapy is STRICTLY CONTRAINDICATED (triggers mania or rapid cycling).



HIGH-YIELD BOARD PEARL: Antidepressant monotherapy is strictly contraindicated in Bipolar Disorder because it can precipitate fulminant manic episodes or rapid cycling. Bipolar I requires only a single lifetime manic episode regardless of whether depressive episodes have occurred.

Topic 3: Major Depressive Disorder (MDD)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Monoaminergic deficiency (serotonin, norepinephrine, dopamine) with neuroendocrine HPA-axis hyperactivity. MDD: at least 5 of 9 symptoms present nearly every day for >= 2 WEEKS, with at least ONE symptom being Depressed Mood OR Anhedonia.

CLINICAL PRESENTATION & PHYSICAL EXAM

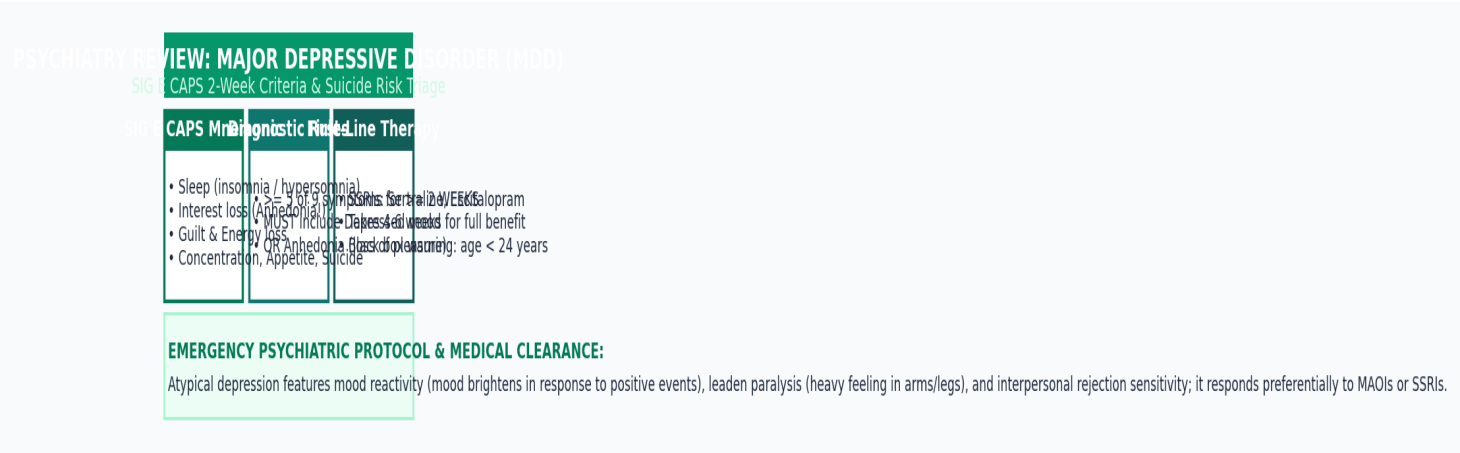
• **Presentation:** Mnemonic "SIG E CAPS": Sleep disturbance (insomnia/hypersomnia), Interest loss (Anhedonia - cardinal!), Guilt or feelings of worthlessness, Energy decrease/fatigue, Concentration loss, Appetite/weight changes, Psychomotor agitation/retardation, Suicidal ideation (active or passive).
• **Physical Exam:** Psychomotor slowing, flattened affect, poor eye contact, tearfulness. Neurologic exam normal.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Clinical DSM-5 criteria. Medical screen: TSH (hypothyroidism is a classic mimic), CBC (anemia), Vitamin B12/Folate, Toxicology screen. Mandatory Suicide Risk Assessment.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Emergency Evaluation: Assess active intent, plan, and means for suicide; if positive, place on 1-to-1 continuous observation and initiate involuntary psychiatric hold. Pharmacotherapy: SSRIs (Sertraline, Fluoxetine, Escitalopram) first line (takes 4-6 weeks for full therapeutic efficacy; warn patient of black box warning for transient increased suicidal ideation in patients < 24 yo). Psychotherapy (CBT).



HIGH-YIELD BOARD PEARL: Diagnosis of Major Depressive Disorder requires at least 2 consecutive weeks of 5 of 9 SIG E CAPS symptoms, and at least ONE symptom must be Depressed Mood or Anhedonia (loss of interest). Black box warning: SSRIs may paradoxically increase suicidal ideation during the initial 2-4 weeks in patients aged under 24 years.

Topic 4: Neurocognitive Disorders: Delirium vs Dementia

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Delirium: ACUTE, fluctuating disturbance in attention, awareness, and baseline cognition triggered by underlying medical illness, medications, or toxins. Dementia (Major Neurocognitive Disorder): insidious, progressive, irreversible cognitive decline with preserved alertness early on.

CLINICAL PRESENTATION & PHYSICAL EXAM

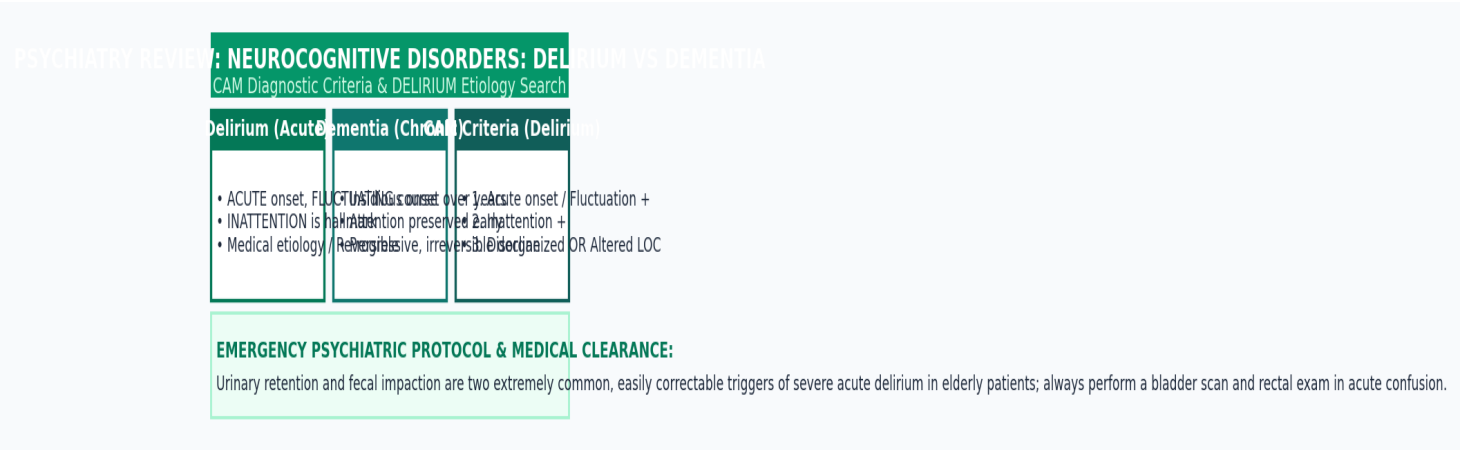
• **Presentation:** Delirium: acute onset over hours/days, waxing and waning course, visual hallucinations, reversed sleep-wake cycle, inattention. Dementia: insidious onset over months/years, memory loss, apraxia, agnosia, alertness intact.
• **Physical Exam:** Delirium: fluctuating sensorium, inattention, myoclonus, autonomic instability, underlying infection signs. Confusion Assessment Method (CAM) positive.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

CAM Algorithm (requires 1 AND 2, plus EITHER 3 or 4): 1) Acute onset & fluctuating course; 2) Inattention; 3) Disorganized thinking; 4) Altered level of consciousness. Comprehensive workup for causes: DELIRIUM mnemonic (Drugs/anticholinergics, Electrolytes, Low O2/hypoxia, Infection/UTI/Pneumonia, Retention [urine/feces], Ictal/post-ictal, Under-hydration, Metabolic). Head CT, UA, CXR, EKG, BMP.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Identify and aggressively treat underlying medical etiology (antibiotics, oxygen, catheterization). Environmental non-pharmacologic interventions (reorientation, family presence, day/night lighting, avoid physical restraints). Pharmacotherapy ONLY for severe agitation threatening self/staff: Low-dose Haloperidol (0.5-1 mg PO/IV) or Quetiapine. BENZODIAZEPINES CONTRAINDICATED (worsens delirium) UNLESS caused by alcohol or sedative withdrawal.



HIGH-YIELD BOARD PEARL: Delirium is an ACUTE, fluctuating disorder of INATTENTION triggered by an underlying medical condition; Benzodiazepines are strictly contraindicated in delirium as they worsen and prolong encephalopathy, UNLESS the delirium is caused by alcohol or sedative withdrawal.

Topic 5: Panic Disorder & Panic Attacks

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Recurrent unexpected panic attacks characterized by an abrupt surge of intense fear reaching a peak within minutes. Panic Disorder requires recurrent unexpected attacks followed by >= 1 month of persistent concern about additional attacks or maladaptive behavior changes (avoidance).

CLINICAL PRESENTATION & PHYSICAL EXAM

• **Presentation:** Abrupt surge of intense fear accompanied by at least 4 of 13 symptoms: palpitations, sweating, trembling, shortness of breath, choking feeling, chest pain/discomfort, nausea, dizziness, chills/heat sensations, paresthesias (perioral/extremity numbness from hyperventilation hypocapnia), derealization/depersonalization, fear of losing control, fear of dying ("impending doom").
• **Physical Exam:** Tachycardia, tachypnea, carpopedal spasm (hyperventilation respiratory alkalosis), diaphoresis, pupillary dilation.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Diagnosis of exclusion. Mandatory EKG (exclude ACS, arrhythmia, WPW, Long QT), Troponin, Blood glucose, Toxicology screen. Differentiate from pheochromocytoma and hyperthyroidism.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Acute Attack in ED: Reassurance in a quiet room, slow diaphragmatic breathing techniques (4-7-8 breathing). Short-acting Benzodiazepine (Alprazolam 0.25-0.5 mg PO or Lorazepam 1 mg PO) for severe acute distress only. Long-term maintenance: SSRIs (Sertraline, Paroxetine) or SNRIs + Cognitive Behavioral Therapy (CBT).



HIGH-YIELD BOARD PEARL: Hyperventilation during a panic attack causes respiratory alkalosis and acute hypocapnia, leading to transient hypocalcemia that manifests as perioral numbness, tingling extremities, and carpopedal spasm (Trousseau sign mimic).

Topic 6: Post-Traumatic Stress Disorder (PTSD) & Acute Stress Disorder

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Trauma- and stressor-related disorders following exposure to actual or threatened death, serious injury, or sexual violence. Acute Stress Disorder (ASD): symptoms persist from 3 DAYS to 1 MONTH after trauma. PTSD: symptoms persist for GREATER THAN 1 MONTH.

CLINICAL PRESENTATION & PHYSICAL EXAM

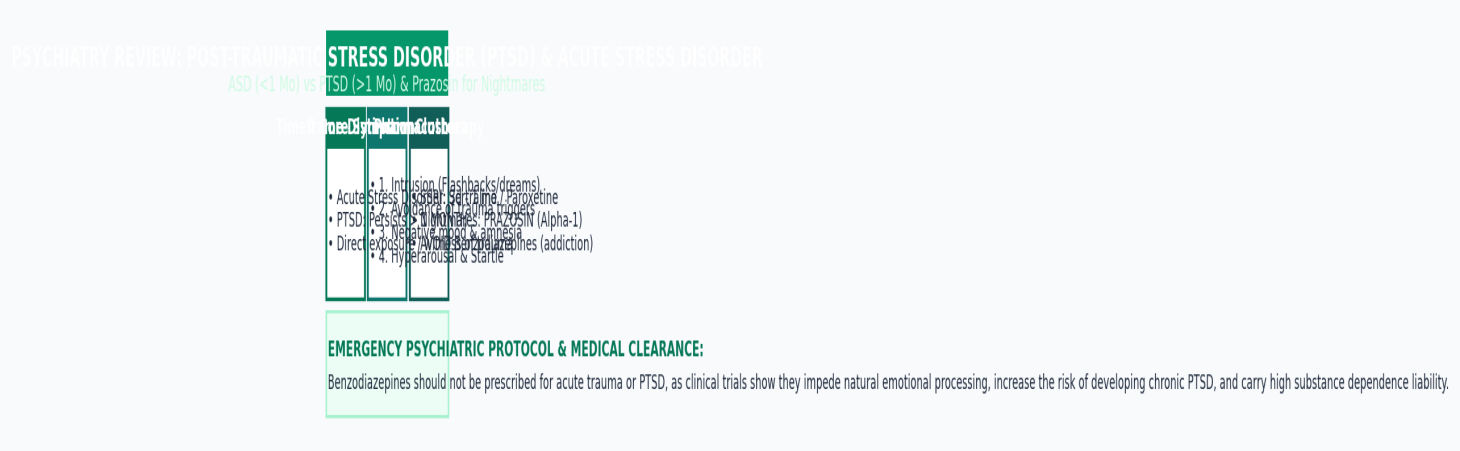
• **Presentation:** Four core symptom clusters: 1) Intrusion (flashbacks, intrusive distressing memories, nightmares); 2) Avoidance (avoiding external reminders, thoughts, feelings); 3) Negative alterations in cognition/mood (amnesia for trauma, detachment, negative beliefs); 4) Arousal and reactivity (hypervigilance, exaggerated startle response, insomnia, irritability).
• **Physical Exam:** Autonomic hyperarousal, tachycardia, hypervigilant behavior, motor restlessness.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Clinical DSM-5 criteria. Assess comorbid major depression, substance use disorders, and suicide risk.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

First-line Psychotherapy: Trauma-focused Cognitive Behavioral Therapy (TF-CBT) or Eye Movement Desensitization and Reprocessing (EMDR). First-line Pharmacotherapy: SSRIs (Sertraline, Paroxetine) or SNRIs (Venlafaxine). Nightmares / Sleep disturbance: PRAZOSIN (alpha-1 adrenergic antagonist; reduces sympathetic outflow during REM sleep). Benzodiazepines are NOT RECOMMENDED (increase risk of PTSD development and substance misuse).



HIGH-YIELD BOARD PEARL: Acute Stress Disorder and PTSD share identical symptom clusters; the sole distinguishing factor is duration: symptoms lasting between 3 days and 1 month represent Acute Stress Disorder, while symptoms persisting for GREATER THAN 1 MONTH represent PTSD. Prazosin is first-line for PTSD-related nightmares.

Topic 7: Schizophrenia Spectrum & Acute Psychosis

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Psychotic disorder characterized by dopaminergic hyperactivity in the mesolimbic pathway (positive symptoms) and hypodopaminergia in the mesocortical pathway (negative symptoms). Timeframe: Brief Psychotic Disorder (< 1 month); Schizophreniform Disorder (1-6 months); Schizophrenia (> 6 months).

CLINICAL PRESENTATION & PHYSICAL EXAM

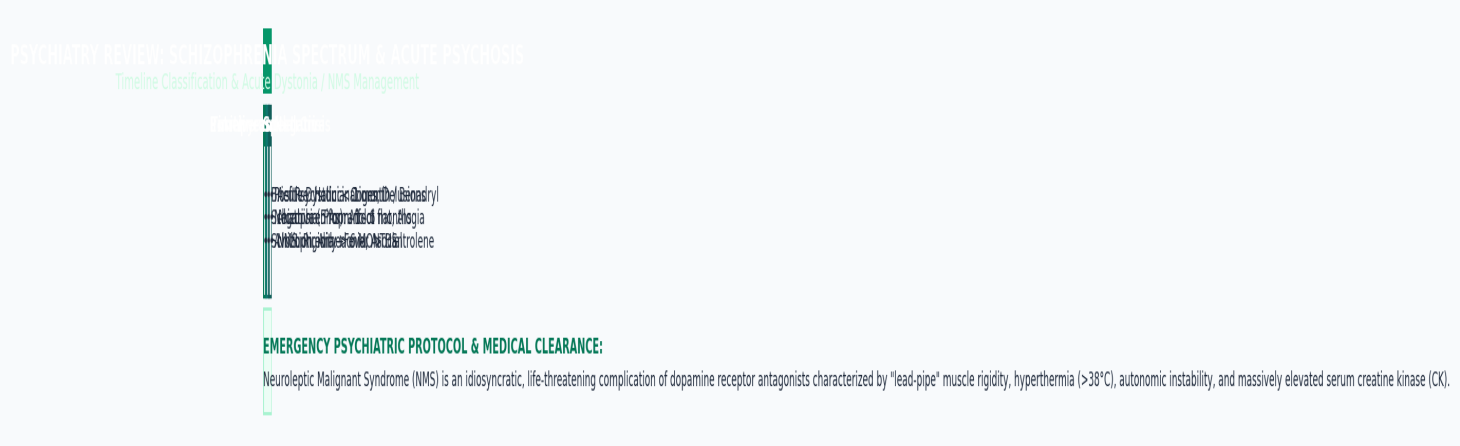
• **Presentation:** Positive Symptoms: Auditory hallucinations (most common), delusions (persecutory, somatic, grandiose), disorganized speech (word salad, loose associations), grossly disorganized or catatonic behavior. Negative Symptoms (the 5 As): Avolition, Alogia (poverty of speech), Affective flattening, Anhedonia, Asociality.
• **Physical Exam:** Disheveled appearance, poor eye contact, blunted affect, paranoid behavior, auditory tracking (listening to voices). Catatonia: waxy flexibility, mutism, echolalia.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Diagnosis of exclusion. STAT Medical Clearance: Toxicology screen (Phencyclidine [PCP], synthetic cannabinoids, amphetamines), CMP (hepatic/renal encephalopathy), Head CT (rule out mass, stroke, bleed), Syphilis serology (RPR), HIV.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Acute Agitation / Psychosis in ED: Second-generation Antipsychotic (Olanzapine 10 mg IM OR Ziprasidone 20 mg IM) OR First-generation Antipsychotic (Haloperidol 5 mg IM) PLUS Lorazepam 2 mg IM. Monitor for Acute Dystonia (treat with Diphenhydramine 50 mg IV or Benztropine 1-2 mg IV). Monitor for Neuroleptic Malignant Syndrome (NMS: lead-pipe rigidity, hyperthermia, autonomic instability, elevated CK; treat with Dantrolene and Bromocriptine).



HIGH-YIELD BOARD PEARL: Acute dystonic reactions (oculogyric crisis, torticollis, laryngeal spasm) can occur within hours to days of starting antipsychotics (especially high-potency haloperidol); treat immediately with IV Diphenhydramine or Benztropine.

Topic 8: Intimate Partner Violence (IPV) & Elder Abuse

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Physical, sexual, or psychological harm by a current or former partner or caregiver. Dynamics involve control, intimidation, isolation, and cycle of violence (tension building -> acute battering -> honeymoon phase).

CLINICAL PRESENTATION & PHYSICAL EXAM

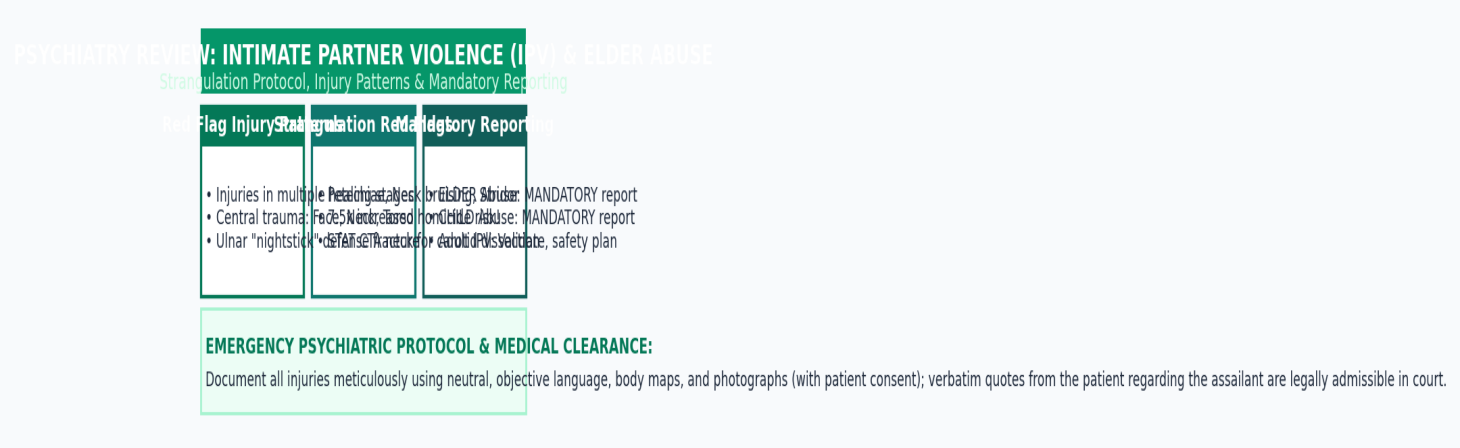
• **Presentation:** Injuries inconsistent with explanation, delayed presentation after injury, frequent ED visits for chronic somatic complaints (headaches, pelvic pain), partner refuses to leave bedside and insists on answering all questions.
• **Physical Exam:** Injuries in varying stages of healing, central injuries (head, neck, chest, abdomen, genitalia; defensive forearm fractures / "nightstick" fractures). In elderly: unexplained fractures, pressure ulcers, severe malnutrition, poor hygiene.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Universal screening in private setting without partner present (HITS questionnaire: Hurt, Insult, Threaten, Scream). Thorough photographic and written medical documentation of all injuries.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Prioritize patient safety: Assess immediate lethality risk (access to firearms, threats of homicide/suicide, strangulation). Strangulation protocol: Soft tissue neck CT / CTA to rule out carotid dissection, laryngeal fractures, and delayed airway edema. Provide confidential domestic violence shelter contact and social work resources. MANDATORY REPORTING: Child abuse and vulnerable elder abuse MUST be reported to Adult Protective Services (APS); adult IPV reporting laws vary by state (respect competent adult autonomy unless deadly weapon used).



HIGH-YIELD BOARD PEARL: Always interview suspected victims of intimate partner violence ALONE in a completely private environment; never question the patient in the presence of the partner. A history of manual strangulation increases the risk of subsequent domestic homicide by 750% and mandates evaluation for carotid dissection and airway edema.

Topic 9: Substance Use Disorders: Overdose & Withdrawal

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

1) Alcohol: Withdrawal triggers CNS hyperactivity (decreased GABA, increased NMDA). Delirium Tremens (DTs) starts 48-96h (confusion, fever, autonomic instability). 2) Opioids: Overdose triggers mu-opioid respiratory depression. 3) Sedatives: Benzo withdrawal can cause fatal seizures. 4) Stimulants (Cocaine/Meth): Sympathomimetic surge.

CLINICAL PRESENTATION & PHYSICAL EXAM

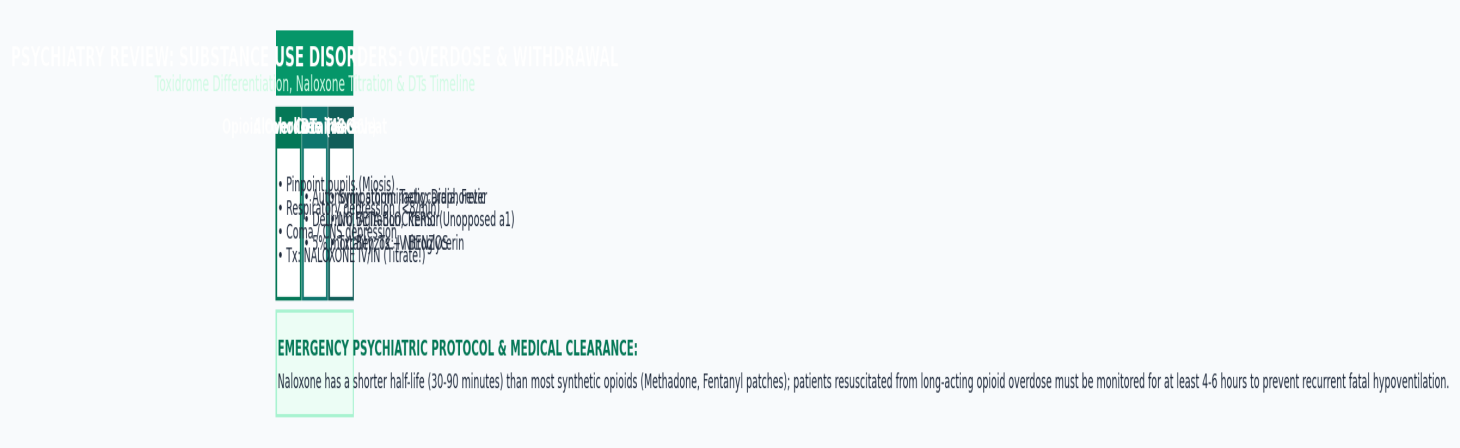
• **Presentation:** Alcohol withdrawal: Tremors, diaphoresis, seizures (6-48h), hallucinations (12-48h, clear sensorium), DTs (48-96h). Opioid overdose: Classic triad of Miosis (pinpoint pupils), Respiratory depression (< 8 breaths/min), and Coma. Cocaine: Chest pain, hyperthermia, mydriasis, hypertension, agitation.
• **Physical Exam:** Opioid: Pinpoint pupils, bradypnea, cyanosis. Sympathomimetic: Dilated pupils, hyperthermia, diaphoresis (distinguishes from anticholinergic dry skin!). DTs: Tachycardia, hypertension, drenched sweat, gross tremor.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Urine drug screen, blood alcohol level, serum acetaminophen/salicylate levels, fingerstick glucose, EKG (check QTc for methadone, wide QRS for cocaine).

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Opioid Overdose: NALOXONE (0.4-2 mg IV/IM/IN; titrate to restore adequate ventilation, not full alertness, to avoid precipitating acute severe withdrawal). Alcohol Withdrawal / DTs: CIWA protocol with high-dose intravenous BENZODIAZEPINES (Diazepam 10-20 mg IV or Lorazepam 2-4 mg IV titrated to light somnolence). Cocaine Chest Pain: IV Benzodiazepines + Nitroglycerin; AVOID BETA-BLOCKERS (unopposed alpha-1 stimulation causes lethal coronary vasospasm and hypertensive crisis!).



HIGH-YIELD BOARD PEARL: Beta-blockers are strictly contraindicated in acute cocaine and amphetamine toxicity because blocking beta-2 vasodilation causes unopposed alpha-1 vasoconstriction, precipitating severe hypertensive crisis and fatal coronary artery spasm. In alcohol withdrawal, Delirium Tremens occurs 48-96 hours after the last drink.

Topic 10: Suicide Risk Stratification & Emergency Management

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Complex biopsychosocial state resulting in self-directed injurious behavior with intent to die. Major risk factors: Previous suicide attempt (SINGLE STRONGEST PREDICTOR), psychiatric illness (MDD, Bipolar, Schizophrenia), substance abuse, access to lethal means (firearms in home), male sex, elderly white males, social isolation, chronic pain.

CLINICAL PRESENTATION & PHYSICAL EXAM

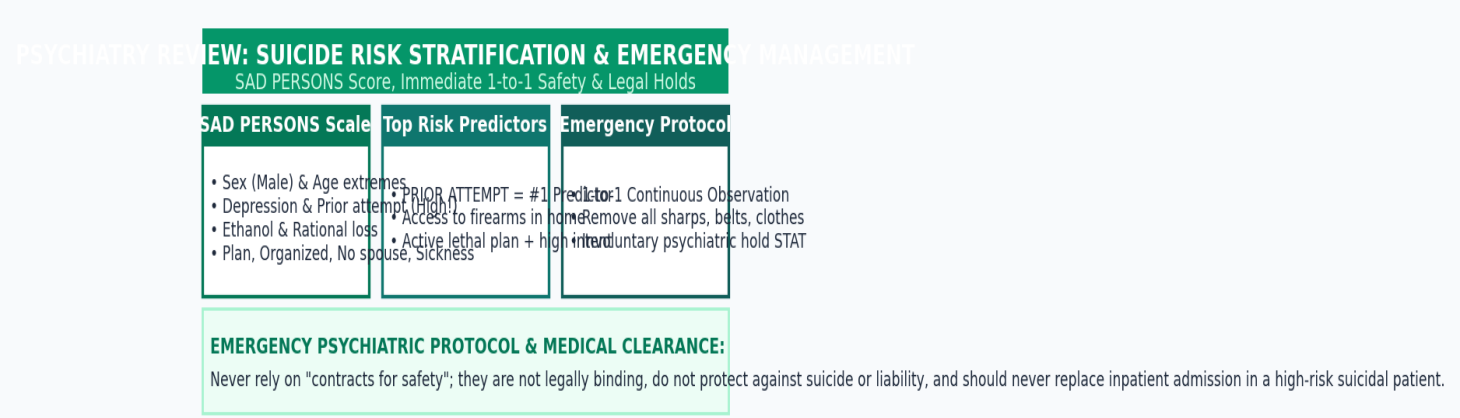
• **Presentation:** Explicit suicidal statements, hopeless ideation, giving away possessions, writing a suicide note, sudden calm demeanor following severe depression (indicates decision made).
• **Physical Exam:** Hesitation marks on wrists/forearms, linear scarring, self-inflicted wounds, signs of intoxication or ingestion.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Structured clinical risk assessment: Columbia-Suicide Severity Rating Scale (C-SSRS) or SAD PERSONS score: Sex (male), Age (<19 or >45), Depression, Previous attempt, Ethanol/substance, Rational thinking loss, Social support lacking, Organized plan, No spouse, Sickness (chronic illness).

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Immediate Safety Protocol: 1) Remove all hazardous objects (belts, shoelaces, sharps); 2) Patient gowns; 3) MANDATORY 1-TO-1 CONTINUOUS IN-PERSON SIGHT-AND-SOUND OBSERVATION; 4) Initiate legal involuntary psychiatric hold if patient refuses voluntary admission. Direct lethal means inquiry: systematically ask about guns, pills, and plans. Inpatient psychiatric hospitalization is mandatory for patients with active intent, specific plan, or lack of protective factors.



HIGH-YIELD BOARD PEARL: A prior suicide attempt is the single strongest clinical predictor of completed suicide. Asking directly about suicidal thoughts, intent, and plans DOES NOT increase the risk of suicide or "plant the idea" in a patient's mind; it is life-saving and mandatory.